Cloud Compute Services Guide

AWS, Microsoft Azure and Google Cloud Platform (GCP)

This guide lists the main compute services offered by the three leading cloud providers and then compares them pairwise: AWS vs Azure, AWS vs GCP, and Azure vs GCP. Service names and features change over time, so check each provider's documentation for the latest details.

Part 1: Compute Services by Provider

1. Amazon Web Services (AWS)

Category	Service	Description
Virtual machines	Amazon EC2	Resizable virtual servers with many instance families (general, compute, memory, GPU).
Simple VPS	Amazon Lightsail	Easy, fixed-price virtual servers for small apps and websites.
Serverless functions	AWS Lambda	Run code on events without managing servers; pay per execution.
Containers	ECS, EKS, Fargate	ECS is AWS-native orchestration, EKS is managed Kubernetes, Fargate runs containers serverlessly.
Batch processing	AWS Batch	Managed batch jobs at any scale.
PaaS	Elastic Beanstalk, App Runner	Deploy apps without managing infrastructure.
Autoscaling	EC2 Auto Scaling	Adjust capacity automatically based on demand.
Hybrid / edge	Outposts, Wavelength, Local Zones	Run AWS compute on-premises or near users.
HPC	ParallelCluster, EC2 HPC instances	High performance computing clusters.
Bare metal / Mac	EC2 Bare Metal, EC2 Mac	Direct hardware access and macOS instances.

2. Microsoft Azure

Category	Service	Description
Virtual machines	Azure Virtual Machines	Windows and Linux VMs in many sizes and series.
Scale sets	Virtual Machine Scale Sets	Deploy and autoscale groups of identical VMs.
Serverless functions	Azure Functions	Event-driven serverless code execution.
Containers	AKS, Container Apps, Container Instances	AKS is managed Kubernetes, Container Apps is serverless containers, ACI runs single containers quickly.
Batch processing	Azure Batch	Large-scale parallel and HPC jobs.
PaaS	Azure App Service	Host web apps and APIs with built-in scaling and CI/CD.
Hybrid / edge	Azure Arc, Azure Stack HCI	Extend Azure management and services to on-premises and other clouds.
Virtual desktops	Azure Virtual Desktop	Cloud-hosted Windows desktops and apps.
HPC	Azure CycleCloud, HPC VMs	Cluster management and HPC-optimised VMs.
Dedicated hosts	Azure Dedicated Host	Physical servers dedicated to your organisation.

3. Google Cloud Platform (GCP)

Category	Service	Description
Virtual machines	Compute Engine	Customisable VMs with predefined and custom machine types.
Instance groups	Managed Instance Groups	Autoscaling and auto-healing groups of VMs.
Serverless functions	Cloud Functions (Cloud Run functions)	Event-driven functions as a service.
Containers	GKE, Cloud Run	GKE is managed Kubernetes (including Autopilot mode); Cloud Run runs stateless containers serverlessly.
Batch processing	Cloud Batch	Managed batch job scheduling and execution.
PaaS	App Engine	Fully managed platform for web apps.
Hybrid / multi-cloud	Google Distributed Cloud, Anthos	Run workloads on-premises, at the edge, or across clouds.
Bare metal	Bare Metal Solution	Dedicated hardware for specialised workloads.
Sole tenancy	Sole-tenant nodes	Physical servers dedicated to your projects.
Specialised hardware	TPUs, GPUs	Accelerators for machine learning workloads.

Equivalent Services at a Glance

Capability	AWS	Azure	GCP
Virtual machines	EC2	Virtual Machines	Compute Engine
Autoscaling	EC2 Auto Scaling	VM Scale Sets	Managed Instance Groups
Serverless functions	Lambda	Functions	Cloud Functions
Managed Kubernetes	EKS	AKS	GKE
Serverless containers	Fargate / App Runner	Container Apps / ACI	Cloud Run
PaaS web hosting	Elastic Beanstalk	App Service	App Engine
Batch	AWS Batch	Azure Batch	Cloud Batch
Hybrid	Outposts	Arc / Stack HCI	Distributed Cloud / Anthos
Bare metal	EC2 Bare Metal	Dedicated Host / BareMetal Infra	Bare Metal Solution

Part 2: Provider Comparisons

AWS vs Azure

Aspect	AWS	Azure
Market position	Largest market share and the broadest service catalogue; earliest to market (2006).	Second largest; strong growth driven by enterprise customers.
VMs	EC2 offers a very large number of instance types, including Graviton (Arm) chips.	Wide VM series; strong Windows Server and SQL Server licensing benefits (Azure Hybrid Benefit).
Serverless	Lambda is mature with a huge ecosystem of triggers and integrations.	Functions has flexible hosting plans and tight integration with Logic Apps and Durable Functions.
Containers	ECS, EKS and Fargate give both AWS-native and Kubernetes options.	AKS has no control-plane fee in its free tier and integrates with Entra ID (Azure AD).
Hybrid	Outposts brings AWS hardware on-premises.	Arc and Azure Stack are generally regarded as stronger hybrid offerings.
Enterprise fit	Strong for cloud-native and startup workloads.	Strong for organisations already using Microsoft 365, Active Directory and Windows.
Pricing	Per-second billing for many instances; Savings Plans and Reserved Instances.	Per-second billing; Reservations, Savings Plans and licence reuse discounts.

Summary: AWS leads in breadth and maturity, while Azure leads in Microsoft ecosystem integration and hybrid cloud.

Choose AWS if: you want the widest range of services, need cloud-native tooling, or have an existing AWS skill base.
Choose Azure if: you rely heavily on Microsoft products, need strong hybrid capabilities, or want licence cost savings for Windows and SQL Server.

AWS vs GCP

Aspect	AWS	GCP
Market position	Largest provider with the most regions and services.	Third largest; known for data, analytics and AI strengths.
VMs	Many fixed instance types and sizes.	Custom machine types let you choose exact vCPU and memory; live migration reduces downtime from host maintenance.
Serverless	Lambda for functions; Fargate for containers.	Cloud Run is widely praised for simple, fast container-based serverless.
Containers	EKS is managed Kubernetes.	GKE, built by the creators of Kubernetes, is considered very polished, with Autopilot mode.
Discounts	Reserved Instances, Savings Plans and Spot Instances.	Sustained use discounts (automatic), committed use discounts and Spot VMs.
AI / ML hardware	Trainium, Inferentia and NVIDIA GPUs.	TPUs and NVIDIA GPUs.
Networking	Large global network with many edge locations.	Premium-tier global private network; global VPC simplifies multi-region design.

Summary: AWS offers breadth and ecosystem depth, while GCP offers simpler pricing, strong Kubernetes and data/AI tooling.

Choose AWS if: you need the broadest service selection, extensive third-party support, or many global regions.

Choose GCP if: you run Kubernetes-heavy, data analytics or machine learning workloads and value automatic discounts and custom VM sizing.

Azure vs GCP

Aspect	Azure	GCP
Market position	Second largest; deeply embedded in enterprise IT.	Third largest; popular with data-driven and developer-focused teams.
VMs	Broad VM series, strong Windows support and licence mobility.	Custom machine types, live migration and sustained use discounts.
Serverless	Functions and Container Apps (built on Kubernetes and KEDA).	Cloud Functions and Cloud Run.
Containers	AKS integrates well with Azure DevOps, GitHub and Entra ID.	GKE has advanced automation (Autopilot, auto-repair, auto-upgrade) and is often seen as the most mature managed Kubernetes.
Hybrid / multi-cloud	Azure Arc manages servers and Kubernetes across environments.	Anthos and Distributed Cloud provide multi-cloud Kubernetes management.
Identity	Entra ID (Azure AD) is the identity backbone for Microsoft shops.	Cloud Identity and IAM, simpler but less tied to enterprise directories.
Pricing	Hybrid Benefit and reservations reduce cost for Microsoft workloads.	Automatic sustained use discounts and committed use discounts.

Summary: Azure is the natural fit for Microsoft-centric enterprises; GCP stands out for Kubernetes, data and AI-focused workloads.

Choose Azure if: your organisation uses Microsoft tooling, needs Active Directory integration, or wants hybrid deployments.
Choose GCP if: you prioritise Kubernetes, big data and ML, or want straightforward automatic pricing discounts.

Quick Decision Guide

Need	Best fit
Widest service catalogue and maturity	AWS
Microsoft ecosystem, Windows and SQL Server licensing, hybrid cloud	Azure
Kubernetes, data analytics, AI/ML, simple discounts	GCP
Serverless containers with minimal setup	GCP (Cloud Run)
Enterprise identity integration	Azure
Startups and cloud-native apps with many third-party integrations	AWS

Note: Pricing, features and service names change frequently. Verify current details on the official AWS, Azure and Google Cloud websites before making decisions.